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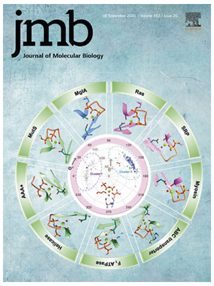

A Distinct Motif in a Prokaryotic Small Ras-Like GTPase Highlights Unifying Features of Walker B Motifs in P-Loop NTPases

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Abstract

A hallmark of the catalytically essential Walker B motif of P-loop NTPases is the presence of an acidic residue (aspartate/glutamate) for efficient Mg^{2+} coordination. Although the Walker B motif has been identified in well-studied examples of P-loop NTPases, its identity is ambiguous in many families, for example, in the prokaryotic small Ras-like GTPase family of MglA. MglA, belonging to TRAFAC class of P-loop NTPases, possesses a threonine at the position equivalent to Walker B aspartate in eukaryotic Ras-like GTPases. To resolve the identity of the Walker B residue in MglA, we carried out a comprehensive analysis of Mg^{2+} coordination on P-loop NTPase structures. Atoms in the octahedral coordination of Mg^{2+} and their interactions comprise a network including water molecules, Walker A, Walker B and switch motifs of P-loop NTPases. Based on the conserved geometry of Mg^{2+} coordination, we confirm that a conserved aspartate functions as the Walker B residue of MglA, and validate it through mutagenesis and biochemical characterization. Location of the newly identified aspartate is spatially equivalent to the Walker B residue of the ASCE division of P-loop NTPases. Furthermore, similar to the allosteric regulation of the Walker B aspartate conformation in MglA, we identify protein families in which large conformational changes involving Walker B motif potentially function as allosteric regulators. The study unravels conserved features of Mg^{2+} coordination among divergent families of P-loop NTPases, especially between ancient Ras-like GTPases and ASCE family of ATPases. The conserved geometric features provide a foundation for design of nucleotide-hydrolyzing enzymes.

Introduction

P-loop NTPase superfamily consists of many important enzymes involved in a variety of cellular processes that require energy. They are among the most ancient and diverse enzyme classes, and comprise more than 10% of all ORFs with known function across all genomes [1]. Some well-known examples of P-loop NTPase superfamily members are small Ras-like GTPases, myosins, kinesins and dyneins, rotary motors such as F_o-F_1 ATPases, and DNA repair and processing enzymes like MutS/MutL, RecA and NTP-dependent restriction enzymes [2,3]. NTPases are enzymes that hydrolyze nucleotide triphosphates (NTP; e.g., ATP or GTP) to diphosphates (ADP or GDP). The superfamily is subdivided into *Kinase-GTPase* and *ASCE* (*a*dditional *s*trand *c*atalytic glutamat*e*) *ATPase* divisions [2–5]. The GTPase division is further divided into *TRAFAC* (*Tra*nslation *Fac*tors) and *SIMIBI* (*Si*gnal recognition particle, *Mi*nD and *Bi*oD) classes. All P-loop NTPases have two signature motifs, the Walker A and Walker B motifs, that are essential for nucleotide binding and hydrolysis [6].

Walker A motif (consensus sequence GXXXXGK[**T/S**]) is essential for binding and orienting the phosphate moiety of NTP, and hence, it is termed as P-loop (phosphate-loop) [7,8]. Hydroxyl group of the threonine or serine in Walker A motif (**T/S** of the motif) usually forms a direct coordination with Mg^{2+} bound to NTP (Figure 1(a)). Walker B motif contains a conserved aspartate or a glutamate (consensus sequence “hhhh[**D/E**]”) that interacts either with a water molecule coordinated to Mg^{2+} (indirect coordination), or occasionally, directly to Mg^{2+} [2,9]. The Walker B acidic residue (**D/E** of the motif) helps maintain Mg^{2+} coordination, which is essential for

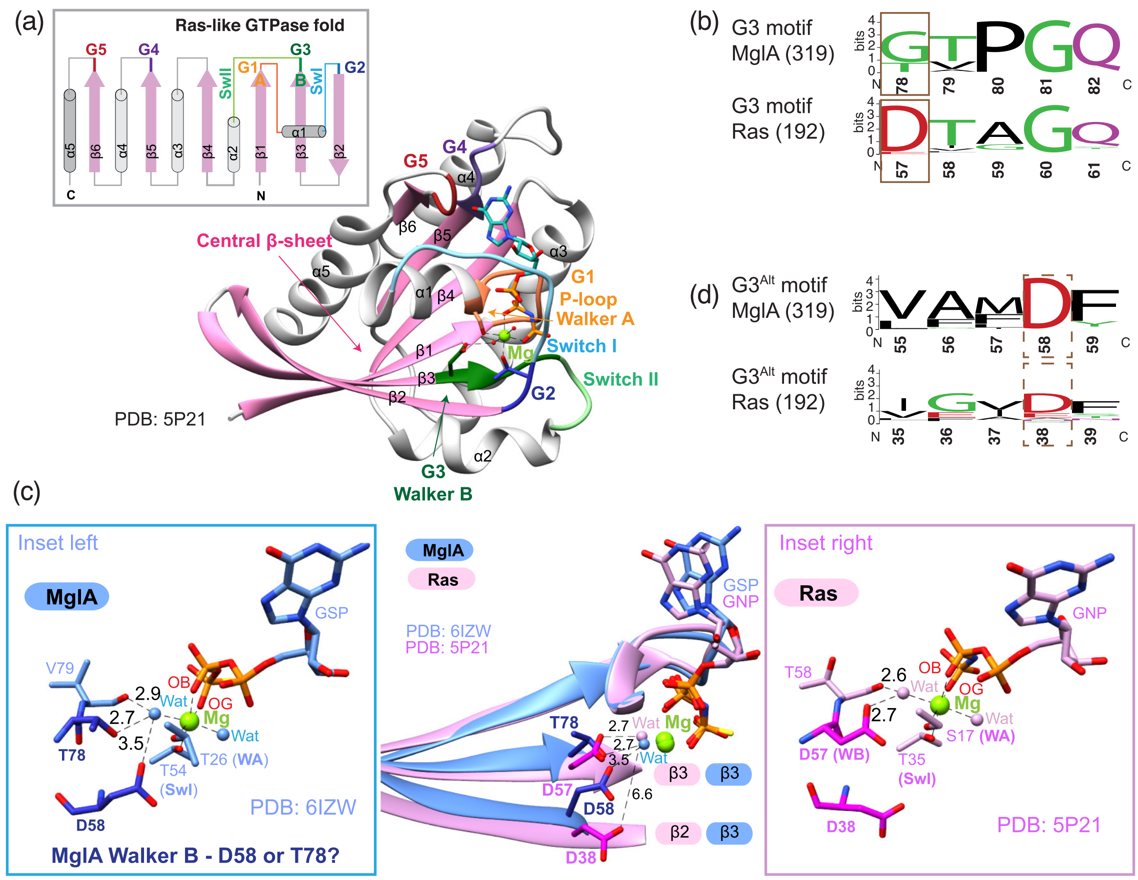


Figure 1. Identification of a Walker B equivalent G3^{Alt} motif in MglA family of prokaryotic small Ras-like GTPases. (a) Topology of small Ras-like GTPases (top). β-strands (pink) and α-helices (gray) are labeled. G1 (orange), G2 (dark blue), G3 (dark green), G4 (purple), G5 (dark red), Switch I (SwI; light blue) and Switch II (SwII; light green) motifs are highlighted in the topology diagram (top) and in the H-Ras structure (below, PDB ID: 5P21) with the central β-sheet shown in pink. (b) Comparison of G3 motifs in prokaryotic MglA family and eukaryotic small Ras-like GTPases. The number of sequences in the alignment used for generating the WebLogo is shown in brackets. The acidic residue in G3 motif (D57) of Ras and the corresponding residue in MglA are highlighted in brown boxes. The residue numbering corresponds to MxMglA (top) and H-Ras (bottom). (c) Structural superposition of MxMglA bound to GTP-γS (blue; PDB ID: 6IZW) and H-Ras bound to GMPPNP (pink; PDB ID: 5P21). Thr78 and Asp58 of MxMglA and Asp57 and Asp38 of H-Ras are shown in sticks (dark blue and magenta, respectively). Green spheres represent Mg^{2+}. Distances, in Å, from the residues to the coordinating water molecules and secondary structure elements are labeled. Insets show MxMglA and H-Ras structures (left and right, respectively) where the residues involved in Mg^{2+} coordination are shown in sticks. Interacting waters (Wat), Switch I (SwI), Walker A (WA), Walker B (WB) and Oβ and Oγ of the nucleotides are highlighted. (d) Comparison of G3^{Alt} motifs in prokaryotic MglA family and eukaryotic small Ras-like GTPases. The number of sequences in the alignment used for generating the WebLogo is shown in brackets. The acidic residue in G3^{Alt} motif (D58) of MglA and the corresponding residue in Ras are highlighted in brown dashed boxes. The residue numbering corresponds to MxMglA (top) and H-Ras (bottom).

nucleotide binding and hydrolysis [10–12]. In most P-loop NTPase families, the hydrophobic residues comprising the motif (hhhh) form a β-strand, with the acidic residue (**D/E**) at its C-terminal tip. Despite Walker B being well defined in many characterized P-loop NTPases, an acidic residue is not present in the canonical motif in some family members. In such cases, the identity of the Walker B residue is ambiguous and requires to be confirmed experimentally.

A prominent family of P-loop NTPases comprises the small Ras-like GTPases. Small Ras-like GTPases belong to the TRAFAC class of P-loop NTPases and act as molecular switches to regulate many signaling processes [13–15]. These GTPases possess five characteristic motifs named G1 to G5 (Figure 1(a)). G1 corresponds to the P-loop (Walker A), G2 comprises the switch I, the aspartate-containing G3 (**D**xxG) is the motif equivalent to Walker B, while G4 and G5 are essential for guanine base specificity [16,17]. The aspartate in G3 motif (**D**xxG) occurs at the C-terminal tip of the β3 strand (Figure 1(a)). The aspartate indirectly coordinates the Mg^{2+} bound to GTP [18] (Figure 1(a)).

A notable example of a P-loop NTPase family without an acidic Walker B residue is the Mutual gliding motility protein A (MglA) family of prokaryotic small Ras-like GTPases. Members of the family have a threonine or a glycine in the G3 motif (residue 78 highlighted in a brown box, G3 motif MglA; Figure 1(b), top) [19] [20] instead of the aspartate which is present in homologous eukaryotic small Ras-like GTPases (residue 57 highlighted in a brown box, G3 motif Ras; Figure 1(b), bottom) [10,11]. MglA family is present in all major phyla of bacteria and archaea, and constitutes an ancestral family of GTPases [20–22]. MglA from *Myxococcus xanthus* (MxMglA) has been extensively characterized for its role in polarity reversals during motility [23,24]. Structures of MxMglA and *Thermus thermophilus* (TtMglA) have been determined [19,25]. Similar to most small Ras-like GTPases, the intrinsic catalytic activity of MglA is stimulated upon binding to its GTPase Activating Protein (GAP). The cognate GAP of MglA is MglB, which functions by orienting the catalytic arginine and glutamine present in MglA (Arg53 on Switch I and Gln82 on Switch II; residue numbering as per MxMglA) to enable GTP hydrolysis [23,24]. One molecule of MglA binds to a dimer of MglB. A major part of the interaction interface between MglA and MglB includes residues from the β2 strand, which undergoes a β-screw movement upon binding either MglB or GTP and its analogs [19,25,26].

The presence of a glycine or threonine in MglA family at the position equivalent to the Walker B acidic residue (highlighted within brown box in G3 motif; Figure 1(b), top) challenges the notion of the requirement of an acidic Walker B residue for Mg^{2+} coordination. Here, a comparison with the crystal structure of H-Ras (human Ras p21) and a comprehensive structural analysis of Mg^{2+} coordination in crystal structures of other P-loop NTPases led us to assign a conserved aspartate (Asp58 in MxMglA located at a strand adjacent to the G3 motif) as the functional Walker B residue in MglA family.

Furthermore, through an extensive structural analysis of nucleotide-bound Mg^{2+} in the Protein Data Bank (PDB), we derive a general rule for unambiguous identification of the functional Walker B residue. This is based on the interaction network of Mg^{2+} in the crystal structure rather than the primary sequence of the protein. Our approach for identifying Walker B residues associated with nucleotide-bound Mg^{2+} provides a method for benchmarking Mg^{2+} sites in P-loop NTPases. The identification of the new functional Walker B aspartate in MxMglA also led us to catalog instances of allosteric regulation in major families of GTPases, including the MglA family, through conformational changes involving the Walker B aspartate. In addition to providing new insights into the catalytic mechanism of P-loop NTPases, our results are of value to understand the evolution and design of this important superfamily of proteins.

Results

$G3^{Alt}$, a new motif in MglA family, contains a conserved aspartate

To identify the functional Walker B residue in MglA, we analyzed the coordination sphere of Mg^{2+} in crystal structures representing the catalytically active conformation of MxMglA. In two of the structures of MglA complexed with a non-hydrolysable analog of GTP (GTPγS) and MglB (PDB IDs: 3T12, 6IZW), the octahedral coordination of Mg^{2+} consists of oxygen atoms from β (OB) and γ (OG) phosphates of GTPγS, Oγ of Thr26 (Walker A) and Thr54 (Switch I) and two water molecules (Figure 1(c), inset left; PDB ID: 6IZW). Oδ2 of Asp58, Oγ of Thr78 and main chain carbonyl atom (O) of Val79 are within 4 Å from one of the waters coordinated to Mg^{2+} (distances labeled in Figure 1(c), inset left). Interestingly, a structural superposition with H-Ras revealed that the side-chains of Asp58 in GTPγS-bound MxMglAB complex (PDB ID: 6IZW) and Asp57 of the G3 motif of H-Ras bound to GppNHp (guanosine-5′-[(β,γ)-imido]triphosphate; PDB ID: 5P21; Figure 1(c)) were spatially close, despite being present on different β strands. This observation prompted us to ask if Asp58 in MxMglA could substitute for the acidic residue seen in most Walker B motifs.

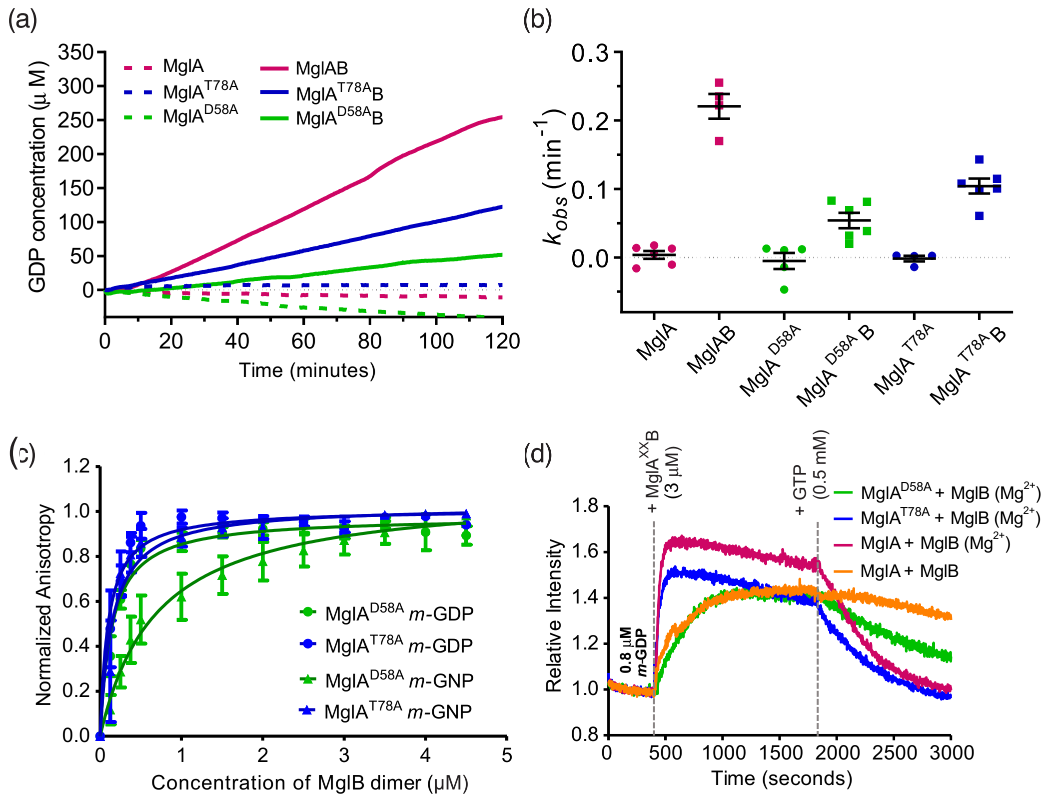
A sequence alignment of the MglA family of prokaryotic small Ras-like GTPases showed that Asp58 is strictly conserved in all classes of MglA family (residue 58 highlighted in a brown dashed line boundary, $G3^{Alt}$ motif MglA; Figure 1(d), top). According to phylogeny, MglA family of proteins has been divided into five classes [20]. The conserved aspartate is part of the consensus sequence “LFF**DF**” (Figure 1(b)) in class I members of the MglA family and “VAM**DF**” in classes II to V [20]. Interestingly, Phe56, Phe57 and Phe59 of the consensus sequence $L^{55}FFDF^{59}$ (residue numbers as in MxMglA) comprise the major interface between MglA and MglB [19,25,26]. These residues form part of the β2 strand that flips upon interaction of MglA with either GTP or MglB [26]. Henceforth, we refer to this novel and highly conserved motif in MglA family that contains Asp58 as $G3^{Alt}$ (for “alternate G3”). The conserved aspartate in $G3^{Alt}$, Asp58, probably performs the function of the canonical G3 motif, namely, coordination of Mg^{2+}, in the MglA family of Ras-like GTPases.

To evaluate the features of Walker B residues and to validate the proposed hypothesis that Asp58 could function as the Walker B aspartate in MxMglA, we used two approaches. To study the roles played by

Asp58 and Thr78, we mutated these residues to alanine in MxMglA and performed GTPase activity measurements and nucleotide binding kinetics. Another approach involved an extensive analysis of the Mg^{2+} coordination in crystal structures of all P-loop NTPases to assess the favored geometry for a Walker B acidic residue interaction.

Asp58 and Thr78 of MxMglA are important for nucleotide binding and hydrolysis

To validate the roles of Asp58 and Thr78 in MglA, we mutated the corresponding residues of MxMglA to alanines. The mutants are referred to as MxMglA^{D58A} and MxMglA^{T78A}, respectively. MxMglA^{D58A} and



Binding affinities of MglB to MglA mutants

	K_D, µM m-GDP	K_D, µM m-GMPPCP
MglA^{D58A}	0.15 ± 0.02	0.70 ± 0.13
MglA^{T78A}	0.10 ± 0.02	0.15 ± 0.03

Nucleotide exchange rates of MglA and mutants in presence of MglB and Mg^{2+}

	k_{on}, µM^{-1}min^{-1} m-GDP	k_{off}, min^{-1} GTP
MglA^{D58A}B	0.29 ± 0.10	0.05 ± 0.01
MglA^{T78A}B	1.35 ± 0.41	0.15 ± 0.11
MglAB	1.12 ± 0.18	0.11 ± 0.01

Figure 2. Biochemical characterization of the mutants MxMglA^{D58A} and MxMglA^{T78A}. (a) GTPase activity measured by the concentration of GDP released (µM) for MxMglA (dark pink), MxMglA^{D58A} (green) and MxMglA^{T78A} (blue). Solid and dotted lines represent the activities in the presence and absence of MxMglB, respectively. Each line represents the average of multiple repeats (at least 3). (b) Comparison of k_{obs} values for MxMglA (dark pink), MxMglA^{D58A} (green) and MxMglA^{T78A} (blue). Circles and squares represent the activities in the absence and presence of MxMglB, respectively. The mean and standard error is shown in black lines (long and short horizontal lines respectively). (c) Fluorescence anisotropy measurements for MxMglB titrated against MxMglA^{D58A} bound to *m*-GDP (green circles) and *m*-GNP (green triangles), and MxMglA^{T78A} bound to *m*-GDP (blue circles) and *m*-GNP (blue triangles). The binding constants are tabulated in the inset below. (d) Kinetics of nucleotide-binding to MxMglA and mutants based on increase in fluorescence of *m*-GDP upon binding to protein. At 400 s, 3 µM MxMglA or mutants with 3 µM MxMglB dimer (point of addition highlighted by a gray line and labeled with "+ MglA^{xx}B 3 µM") were added to 0.8 µM solution of *m*-GDP. Bound *m*-GDP was competed out by adding excess of unlabeled GTP at 1800 s (point of addition marked by a gray line and labeled with "+ GTP 0.5 mM"). The rates of association and dissociation are tabulated below.

MxMglA^{T78A} were well folded and monomeric (as deciphered from size exclusion chromatography) (Supplementary Figure 1a). The purified mutants contained bound GDP, despite GDP not being added during any step of purification. This was observed from ion exchange chromatography profiles of the protein extracts, which exhibited a prominent peak corresponding to GDP elution (Supplementary Figure 1b).

We measured the rate of GTP hydrolysis of the mutants by monitoring the rate of GDP released using NADH coupled enzyme assay (Figure 2(a)). In the presence of MxMglB, both MxMglA^{D58A} and MxMglA^{T78A} mutants showed 2- to 4-fold lower GTPase activity (estimated by k_{obs}, the concentration of GDP released per minute per μM concentration of the GTPase) compared to that of MxMglA (Figure 2(b)). Earlier studies have also reported that Walker B residue mutants retain GTPase activity, though at a lower rate compared to the wild-type. For example, biochemical studies for H-Ras D57A (Walker B residue mutant) showed only a 2-fold decrease in GTP hydrolysis [10].

The decrease in GTP hydrolysis rate could be due to the effect of the mutations on binding MglB and/or GTP. We have shown previously that MxMglB binds MxMglA in the presence of either GDP or GTP [25]. The binding with MxMglB in the presence of GDP or GTP indicates that the mutants are capable of undergoing the β-screw movement required for formation of binding interface [25]. Both mutants were capable of binding to MxMglB in presence of either GDP or GTP (Figure 2(c)). To find if the mutations affected the binding affinities for MxMglB, we performed fluorescence anisotropy measurements of 2′/3′-*O-(N-methyl anthraniloyl (mant*)-labeled nucleotide bound to MxMglA mutants upon titration with increasing amounts of MxMglB. The affinities of MxMglB for *mant*-GDP bound state of MxMglA^{D58A} and MxMglA^{T78A} (0.15 (±0.02) μM and 0.10 (±0.02) μM, respectively), were comparable to the reported value for MxMglA (0.1 (±0.01) μM; Figure 2(c); [25]). However, the affinities of MxMglB to the corresponding *mant*-GMPPCP bound states were 0.70 (±0.13) μM and 0.15 (±0.03) μM for MxMglA^{D58A} and MxMglA^{T78A}, respectively (Figure 2(c)).

Next, we monitored the kinetics of nucleotide exchange for a mix of MxMglA with an equimolar ratio of MxMglB dimer, by measuring fluorescence changes upon binding of *mant*-GDP, followed by the decrease in fluorescence upon addition of excess of GTP. The rates of association of *mant*-GDP for the mutants (k_{on}), and the rates of dissociation of *mant*-GDP upon addition of GTP (k_{off}) indicate their ability to bind GDP and GTP, respectively. Because both Asp58 and Thr78 interact with the water coordinated to Mg^{2+} only in the conformation bound to MglB (Supplementary Figure 1c), we compared the k_{on} and k_{off} rates for the mutant and wild-type proteins in the presence of MxMglB. To understand the extent of requirement of Mg^{2+}, we compared the rates of nucleotide exchange in the presence of MxMglB for the wild-type MxMglA with and without Mg^{2+}.

In the absence of Mg^{2+}, MxMglA (wild-type) showed markedly slower rate of association of *mant*-GDP and very slow exchange with GTP (Figure 2(d), Supplementary Figure 1d). The binding kinetics of MxMglA-^{D58A} with *mant*-GDP in the presence of Mg^{2+} matched closely with that of wild-type MxMglA in the absence of Mg^{2+}, suggesting that Mg^{2+} coordination is affected in the mutant (Figure 2(d), Supplementary Figure 1d). For MxMglA^{T78A}, k_{on} for *mant*-GDP binding was comparable to that of the wild-type while k_{off} for *mant*-GDP release upon addition of excess GTP was faster (Figure 2(d), Supplementary Figure 1d). Hence, mutation of Asp58 appeared to have a more pronounced effect on Mg^{2+}-dependent exchange kinetics of nucleotides compared to that of Thr78 (Figure 2(d)). Therefore, we concluded that Asp58 of G3^{Alt} motif plays the role of Walker B residue rather than Thr78 of G3 motif.

Dataset of P-loop NTPase structures for identification of Walker B residues

Based on mutational analysis, we confirmed that Asp58 functions as the Walker B residue for the MglA family. This prompted us to analyze Mg^{2+} coordination in all P-loop NTPases to identify characteristic features of a Walker B residue. We analyzed the atoms coordinating Mg^{2+} bound to nucleotides (see Supplementary Table 1 for list of nucleotides) in a database of non-redundant protein structures from the PDB, resolved to better than 3.0 Å and R_{free} less than 30% (Figure 3(a), boxes I and II; refer to Methods for further details). The atoms coordinating Mg^{2+} were given a positional description (Figure 3(a), boxes III and IV). Positions 1 and 2 denoted OB and OG atoms of the nucleotide, respectively. These positions fixed the orientation for assigning positions 3 to 6 to the other four coordinating atoms (Figure 3(a), box III). To assign positions 3 to 6, we calculated nine pairwise angles between the six atoms, two of which are expected to be ~180° and the rest ~90° (Figure 3(a), box III). Assuming positions 1 to 4 (numbered clockwise) to lie in an equatorial (horizontal) plane, positions 5 and 6 denoted coordinating atoms above and below the plane, respectively (Figure 3(a), box IV).

Structures with relative angles deviating from the expected 180° and 90° values (range of 150 to 180° and 90° (±20°)) were excluded from the analysis. This criterion enabled us to eliminate structures with incorrect geometry of Mg^{2+} ion coordination (Figure 3(a)). Thus, the approach provides a powerful means for benchmarking Mg^{2+} sites in P-loop NTPases. There are many structures in the PDB with incorrectly modeled Mg^{2+} ions, which possess an inaccurate geometry of Mg^{2+} coordination. This facile method complements existing resources for ensuring correct placement of metal ions with accurate coordination

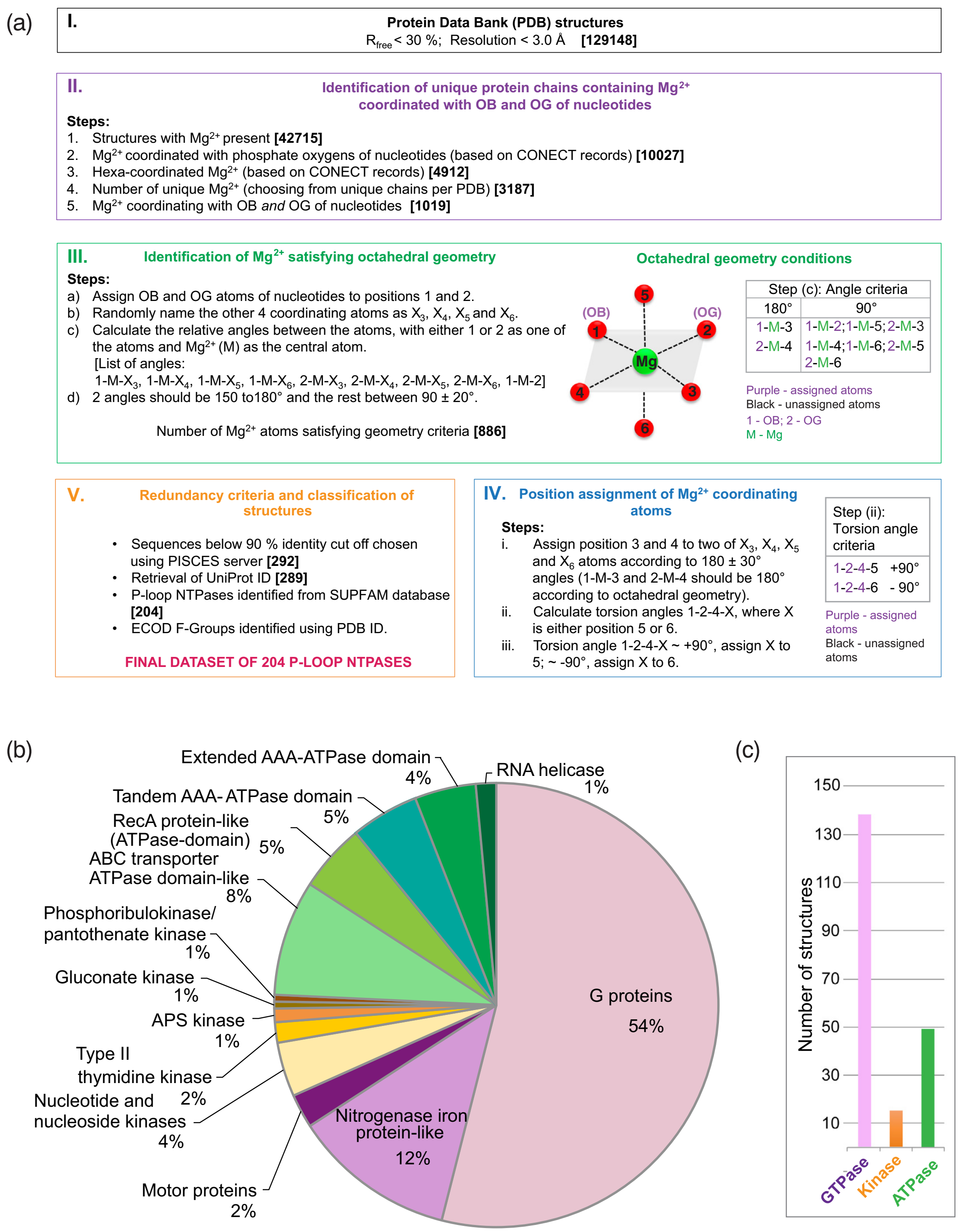


Figure 3. Dataset of P-loop NTPase structures for identification of Walker B residues. (a) Flowchart showing the steps for shortlisting structures of P-loop NTPases based on Mg^{2+} coordination to determine structural features defining a Walker B residue. The numbers of structures at each step is given within square brackets. (b) Family-wise distribution of P-loop NTPases in the dataset. (c) Bar diagrams showing the distribution of total number of structures across GTPase, Kinase and ATPase divisions.

geometry during model building of structures [27,28].

The method based on position assignment provided us with a final dataset of validated Mg^{2+} sites in 204 structures belonging to P-loop NTPase families and 65 of other families (non-P-loop NTPases) (Figure 3(a); Supplementary Tables 2 and 3). We proceeded with further analysis of only the P-loop NTPases (Figure 3(b)). Though representatives from all major divisions of P-loop NTPases were present in the dataset (Figure 3(c)), family-wise distribution of the P-loop NTPases reflected a larger representation of G-proteins (Figure 3(b) and (c)). In the evolutionary classification of proteins as per the ECOD database [29], P-loop NTPase structures are distributed into 202 F-groups. P-loop NTPase structures containing both NTP and Mg^{2+} ligands and those qualifying the redundancy and quality criteria (sequence identity cutoff 90%; resolution better than 3.0 Å and R_{free} less than 30%) belonged to only 69 of these F-groups. Among these 69 F-groups, 51 F-groups were represented in our dataset while 18 F-groups were not represented probably due to lack of structures that qualify the Mg^{2+} coordination geometry criteria. Supplementary Table 4 shows the list of F-groups among P-loop NTPases and their representation in our dataset.

Residue-wise distribution and motifs of Mg^{2+} coordination

Next we mapped the positions of coordinating atoms to protein motifs in P-loop NTPases. With the eukaryotic small Ras-like GTPase (H-Ras; PDB ID 5P21) as reference, positions 3, 4 and 6 corresponded, respectively, to Oγ of threonine residue in Switch I loop, Oγ of Walker A threonine/serine, and the water molecule held by the Walker B aspartate (Figure 4 (a)). Thus, residues interacting with the water molecule at position 6 (indirect Mg^{2+} coordination) or residues directly interacting with Mg^{2+} at position 6 were potential candidates for being part of the Walker B motif.

A position-specific distribution of coordinating residues or water molecules in P-loop NTPases (Figure 4(b)) showed that positions 3 and 4 consisted mostly direct contacts by protein atoms (111 and 193, respectively). The prominent types of direct contacts at position 3 belonged to Oγ1 of Switch I threonine in small Ras-like GTPases (Figure 4(a)) [19,25,26], Oγ of Switch I serine in motor proteins of P-loop GTPase division [30], and Oε2 of the Q-loop glutamine in ABC transporter families [31]. Direct contacts at position 4 comprised Walker A residue contacts (77 and 114 contacts of Ser and Thr, respectively) in all P-loop NTPases (Figure 4(a) and (b); see Supplementary Table 5 for exceptions).

Unlike positions 3 and 4, water molecules formed a large proportion of the coordinating atoms at positions 5 (all 204) and 6 (198 out of 204) in P-loop NTPases. High occurrence of Asp/Glu among direct and indirect contacts at position 6 (Supplementary Tables 6 and 7) highlights the requirement of an acidic residue for Mg^{2+} coordination in P-loop NTPases. For most exceptions without an acidic residue contact (Supplementary Tables 7 and 8), an aspartate corresponding to Walker B position was present in the vicinity, but above 4 Å distance cut off. These structures might have been captured in inactive conformations, and probably form a contact mediated by Asp/Glu in an active conformation.

Features of interactions with position 6 water molecule

To understand the spatial geometry of an indirect interaction by a Walker B residue, the location of protein atoms interacting with position 6 water (labeled as 6′ in Figure 4(a)) was mapped. Three parameters were calculated: i) distance "*d*" between atoms 6 and 6′, ii) the angle "*θ* " formed by Mg^{2+}, position 6 water molecule and 6′ atom, and iii) torsion angle "*φ*" subtended by the four atoms, namely, position 4 atom, Mg^{2+}, position 6 water molecule and 6′ atom (see Figure 4(a) and Methods for definition). The residue-wise distribution of distance *d* for all the atoms at position 6′ is shown in Figure 4(c). Indirect contacts at position 6 (atom 6′) present in P-loop family of NTPases showed two clear clusters in the spatial distribution (θ–φ plot; Figure 4(d)). Cluster 1, predominantly comprising aspartates, populated around 75° < θ > 150° and −30° < φ > 60° (−30° corresponds to 330°; cluster labeled within purple boundary in Figure 4(d)). Cluster 2 was around 75° < θ > 150° and 110° < φ > 180° (labeled within cyan boundary in Figure 4d).

Manual investigation of the aspartates in cluster 1 identified most of them to be part of the canonical Walker B motif of the respective proteins. Cluster 2 mainly consisted of main chain O in proteins of the GTPase family (e.g., Thr58 in H-Ras; PDB ID: 5P21; Figure 1(c)), and side-chain carboxyl oxygen of glutamate/aspartate in proteins of the ASCE family (e.g., Glu694 in MutS; PDB ID: 1W7A). In 58 structures with only 1 contact for the position 6 water, the single contact corresponded to an aspartate located in cluster 1 of the θ–φ plot (Supplementary Table 7).

Interactions between motifs at position 6 water molecule

In most structures in which position 6 water had two contacts, the protein residue in cluster 2 was adjacent (according to primary sequence) to the residue in cluster 1 (Figure 5(a); see Supplementary Table 9 for exceptions). For ASCE ATPase family members, these residues corresponded to the catalytic glutamate (or aspartate in a few examples) (Figure 5(a)). An exception is the cystic fibrosis conductance regulator (CFTR) family (PDB ID: 1R0X), which possesses a serine in cluster 2 interaction (Figure 5(a)).

Next, we inspected examples in which contacts for position 6 water were not aspartate or glutamate in cluster 1 or in which cluster 1 and 2 residues were not adjacent residues. MxMglA-Asp58 was located

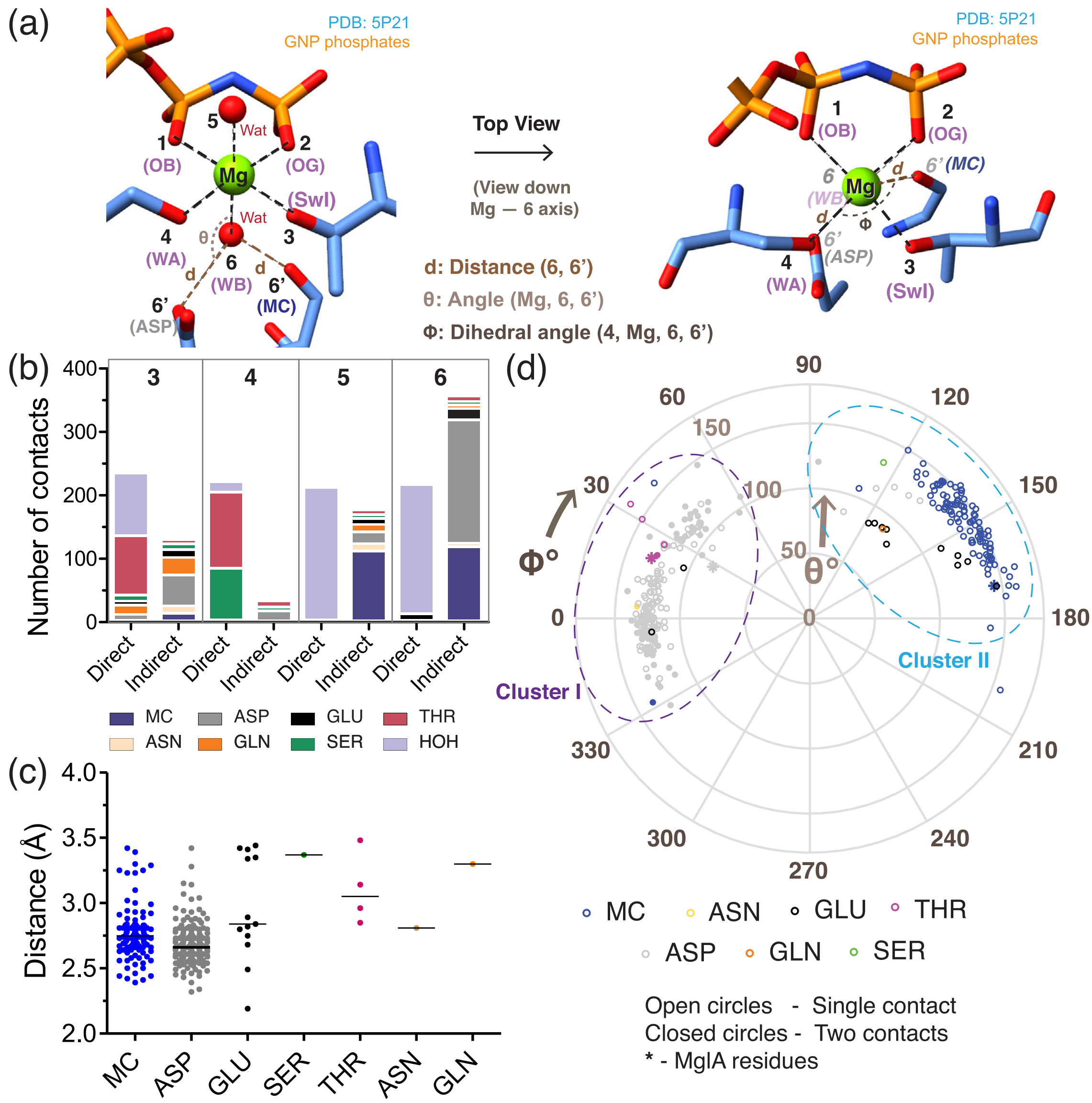


Figure 4. Features of the interaction network of Mg^{2+} coordination in P-loop NTPases. (a) Positional description of coordination in nucleotide-bound Mg^{2+}. Positions 1 and 2 are fixed as the oxygen atoms of β- and γ-phosphates (labeled as OB and OG) of the nucleotide that interacts with Mg^{2+}, respectively. 3 and 4 are the positions assigned clockwise in the same plane defined by OB-Mg-OG, while 5 and 6 are perpendicular (above and below, respectively) to the plane. Definition of the position-wise coordination is illustrated using H-Ras (PDB ID: 5P21) as an example. Angle θ and dihedral angle φ are depicted on the figure in the side view and top view, respectively. The multiple (two in this figure) contacts to the atom at position 6 are labeled as 6′. The interacting motifs WA, WB, SwI represent Walker A, Walker B and Switch I, respectively. ASP and MC denote aspartate residue and main chain, respectively. In the top view, atoms shown below the plane are labeled in italicized font (6, 6′, WB, ASP and MC). Position 6 water is behind the sphere representing Mg^{2+}, while, for clarity, position 5 water is not shown. (b) Position-wise distribution of direct and indirect contacts in P-loop NTPases. Gaps were introduced between bar segments of individual residues to improve clarity. The numbers labeled on top of each box denote the contact positions. Amino acids (represented in their three-letter code) are color coded according to the legend below. (c) Distribution of distances between indirect coordinating atom and water molecule at position 6 in P-loop NTPases. Mean values for each residue are highlighted by horizontal lines. (d) Spatial preference of residues at interacting with water molecule at position 6 in a polar plot representation of angle *θ* and dihedral angle *φ* (both in degrees) for P-loop NTPases. Closed and open circles depict data points from structures with one and two contacts, respectively, while "*" highlights the residues from MxMglA structure (PDB ID: 6IZW). "MC" denotes main chain carbonyl oxygen, and the color code for residues are shown.

in cluster 1. Thr78 of MxMglA and TtMglA were two among the five threonines located at the narrow range of φ from 20 to 35° within cluster 1 (Figure 4(d)). The three other structures containing threonine contacts at position 6 were from the ASCE family members namely, SV40 helicase (PDB ID: 1SVM), RadA recombinase (PDB ID: 3NTU) and NSF (PDB ID: 1D2N). All three also possessed a Walker B acidic residue (Figure 5(b)). Superposition of these ASCE family members on to MxMglA showed that the threonine was structurally equivalent to MxMglA Thr78 (Figure 5(b)). This led us to conclude that threonine probably facilitates Mg^{2+} coordination in these ASCE family members.

The observation of a threonine, which plays a role in complementing a Walker B acidic residue interaction in ASCE family members, prompted us to look for conservation of the threonine in other members of P-loop NTPases. In addition to the above examples, PilB (PDB IDs: 5TSG, 5TSH), PilF (PDB ID: 5OIU), KaiC (PDB ID: 4TL6) and terminase (Ser instead of Thr; PDB ID: 4IEE) were the other ASCE family members having a conserved serine or threonine at a structurally equivalent position compared to Thr78 in MxMglA. However, the threonine residues were at 3.8–4.5 Å from the water at position 6 in the captured conformations in the crystal structures (Figure 5(b)). Interestingly, the threonine is within 3.5 Å distance of the side-chain of Walker B aspartate (atom interacting with position 6) in all these structures (Figure 5(b)), suggesting that the threonine might play a role in orienting the Walker B acidic residue.

Analysis of the structural features of Walker B aspartate/glutamates also revealed that their side-chain atoms interacting with the position 6 water were within hydrogen bonding distance of the Oγ atom of Walker A threonine/serine in majority of the cases (Figure 5(a)). The hydrogen bond appeared to hold the side-chains of Walker A and B residues in an orientation that enabled coordination with Mg^{2+}. This is consistent with the observation that mutation of Walker B aspartate to asparagine in MutS affected the orientation of Walker A threonine [32]. We conclude that the function of optimal orientation of Walker A cannot be performed by a threonine in the adjacent strand or any other polar residue substituting a Walker B acidic residue.

Location of MxMglA-Asp58 is spatially similar to the Walker B residue of ASCE family NTPases

Having identified the Walker B aspartate in MxMglA, we proceeded to compare the location of MxMglA Asp58 residue with those of bona fide Walker B residues in structures of other P-loop GTPases and ATPases. P-loop GTPases and ATPases are divided into three divisions, namely, TRAFAC, SIMIBI and ASCE, having a conserved core of a central β-sheet sandwiched by α-helices (Figure 6(a)). We superposed a set of representative structures from each sub-family of P-loop GTPases and ATPases (Tables 1 and 2; [2–4,33]). We adopted a uniform numbering scheme for the relevant strands in the core β-sheet to facilitate comparison of the varied structural folds (Figure 6(a); see Methods for details). According to the scheme, strand preceding the P-loop was considered as 1, and rest of the strands numbered in the order of their occurrence in the central β-sheet. Each strand was assigned either "P" or "A" depending on whether they were parallel or antiparallel with respect to its preceding strand (Figure 6(a)).

Among members of TRAFAC and SIMIBI classes, the bona fide Walker B residue is usually present in strand 2P (according to our nomenclature) next to the β-strand preceding the P-loop (labeled as strand 1P) (Figure 6(a) and (b), Table 1). In the ASCE division of P-loop ATPases, the Walker B residue is present in strand 3P (Figure 6(a) and (c), Table 2). In comparison, MxMglA Asp58 is located on strand 3A similar to the position of Walker B residue on strand 3P of ASCE (Figure 6(b) and (c)). As emphasized by the naming scheme of the strands, strand 3A in MglA is antiparallel with respect to the other strands of the β-sheet, while strand 3P is parallel in ASCE members (Figure 6(c)).

PilB/PilT subfamily of ATPases was the sole exception in which the Walker B residue was absent from the central β-sheet [34]. In this family, a glutamate, located on a loop, interacts with water molecules coordinating the Mg^{2+} at positions 3 and 5 (Figure 6(d)). A general trend, especially among P-loop GTPase division, appeared to be that aspartates at the Walker B position form indirect contacts, while glutamates form direct contacts (Figure 6(e); Table 1). All examples of direct contacts of Walker B residue among members of P-loop GTPase division in our analysis (both from Mg^{+2} coordination identified from PDB analysis and from the representative structures in Tables 1 and 2) were glutamates. Among ASCE ATPases in our representative dataset, we identified two examples of indirect contact by glutamate. These included GP2 terminase of bacterial virus SF2 (PDB ID: 4IEE, Figure 6(f)) and SV40 helicase (PDB ID: 1SVM, Figure 5(b)). However, the Mg^{2+} coordination geometry in the terminase structure was distorted and did not satisfy our geometry criteria (Figure 3(a), box IV). There were no examples of direct contacts by aspartates at position 6 among P-loop NTPase structures satisfying the Mg^{2+} coordination criteria (Figure 3(a)) even in the extended set of structures before applying redundancy filter, indicating that this is a rare occurrence among P-loop NTPases.

Walker B residue as an allosteric regulator

A special feature of the aspartate (Asp58) in the $G3^{Alt}$ motif of the MglA family (the proposed Walker

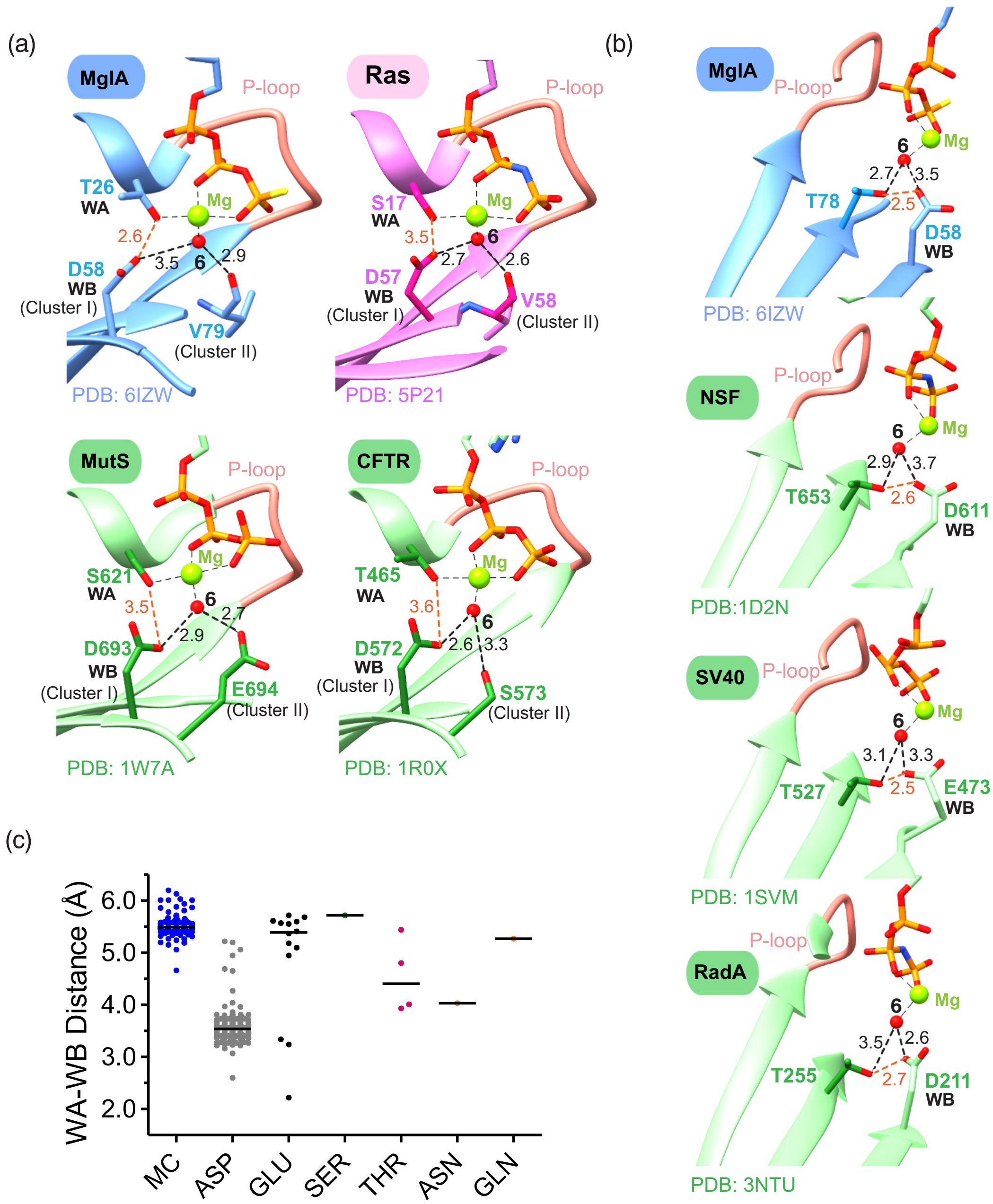


Figure 5. Interacting motifs at position 6 water molecule in P-loop NTPases. (a) Representative structures possessing two contacts with position 6 water. The interacting residues belonging to cluster 1 and cluster 2 in MxMglA (blue; PDB ID: 6IZW), H-Ras (pink; PDB ID: 5P21), MutS (green; PDB ID: 1W7A) and CFTR (green; PDB ID: 1R0X) are shown. ASCE family members are shown in green, TRAFAC (H-Ras) in pink and MxMglA in blue. The structures are shown in the same orientation after superposition based on their P-loop residues. In each sub-panel, the distances between position 6 water and cluster I and II residues are marked by thick black dotted lines, and the distance between Walker A (WA) Ser/Thr Oγ and Walker B (WB) atom that interacts with position 6 water is marked by an orange dotted line. The corresponding distances (in Å) are labeled. Mg^{2+} and position 6 water are shown by green and red spheres, respectively. (b) Representative ASCE members containing threonine equivalent to MxMglA Thr78. The corresponding threonines are highlighted in darker shades and their interacting distances with position 6 water (thick black dotted lines) and Walker B atom (residue labeled as WB, orange dotted lines) marked and distances labeled. MxMglA (PDB ID: 6IZW; blue), NSF (PDB ID: 1D2N; green), SV40 helicase (PDB ID: 1SVM; green) and RadA recombinase (PDB ID: 3NTU; green) are shown in the same orientation after superposition based on their P-loop residues. The corresponding distances (in Å) are labeled. Mg^{2+} and position 6 water are shown by green and red spheres, respectively. (c) Distribution of distances of indirect coordinating atom and water molecule at position 6 in P-loop NTPases. Mean values for each residue are highlighted in the plot.

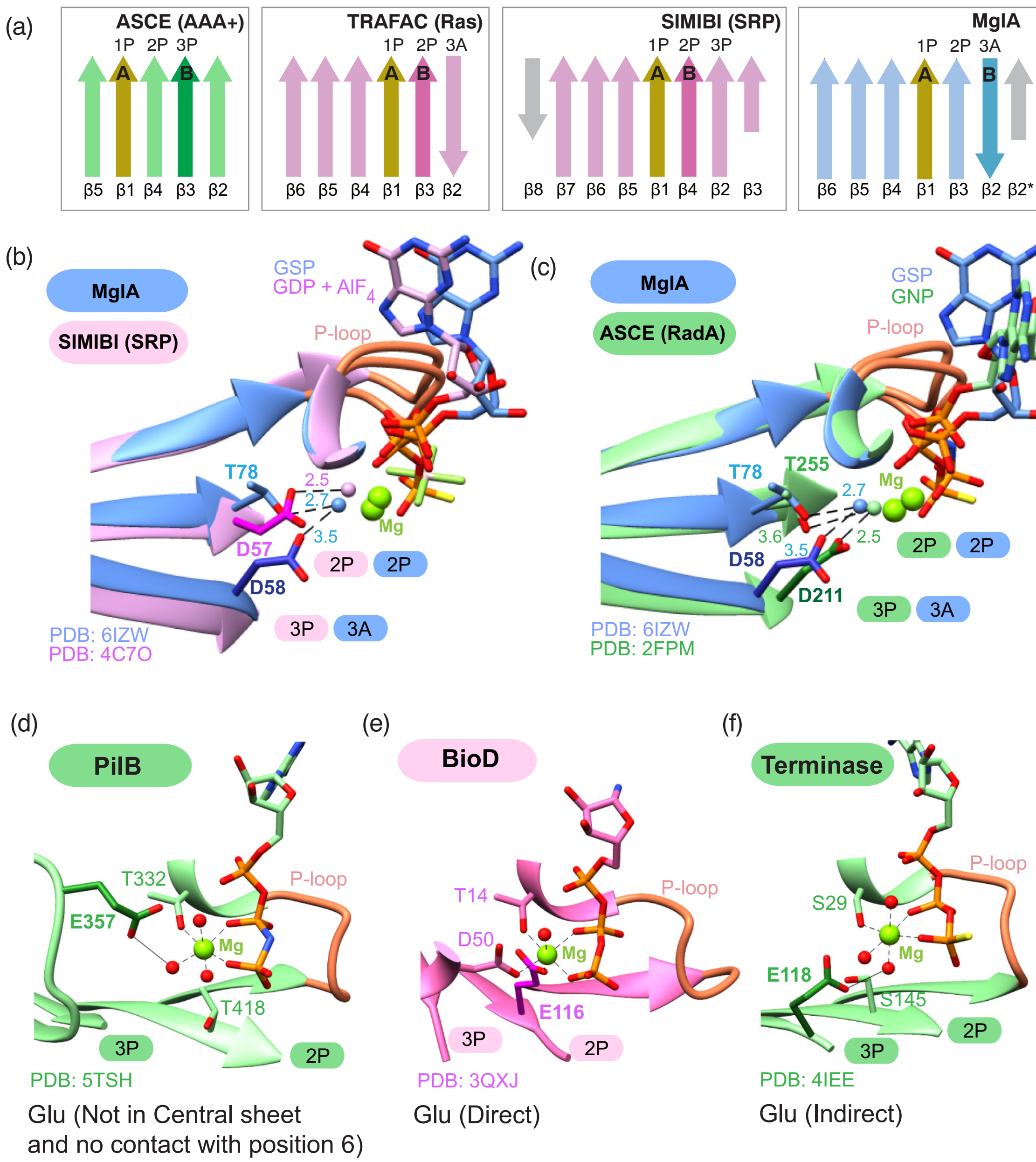


Figure 6. Spatial location of Walker B in P-loop NTPases. (a) Central β sheet topology diagram of major P-loop NTPase families. ASCE (green; representative: AAA+), TRAFAC (pink; representative H-Ras), SIMIBI (pink; representative: SRP) and MglA (blue), with Walker A and Walker B related strands highlighted as A and B, respectively. Strands are labeled according to the nomenclature of the fold (below) and our nomenclature (above). (b) Superposition of MglA (blue; PDB: 6IZW) and SIMIBI (representative SRP; pink, PDB: 4C7O). Asp58 (sticks; dark blue) of MglA is present in strand 3A, while Asp190 (sticks; purple) of SRP is present in strand 2P. Distances between residues and position 6 water labeled in black dotted lines. (c) Superposition of MglA (blue; PDB: 6IZW) and ASCE (representative RadA; green; PDB: 2FPM). Asp211 of RadA (sticks; dark green) is present in strand 3P. (d–f) Variation in residues for magnesium coordination in P-loop NTPases. Residues involved in Mg^{2+} coordination are shown in stick representation. Walker B residue is highlighted in darker shade (dark green or purple). Examples of coordination by glutamate outside the central β sheet and no contact with position 6 water (PilB; PDB ID: 5TSH; (d)), direct coordination by glutamate (BioD; PDB ID: 3QXJ; (e)) and indirect coordination by glutamate (terminase; PDB ID: 4IEE; (f)) are shown.

Table 1. List of representative structures of P-loop GTPases

Superfamily	Family	Protein Name	PDB ID	Resolution (Å)	Chain ID	Nucleotide	Walker B D/E	Walker B strand number	Walker B strand orientation	Mg^{2+} coordination (Direct/indirect)	P-loop residues (CA superposed)
Division: P-loop GTPases; Class: TRAFAC											
Translation factor superfamily	Classical Translation Factor	*E. coli* EF-Tu	2BVN	2.3	B	GNP	D80	2	Parallel	Indirect	14-18,25-27
		E. coli EF-Tu	1DG1	2.5	G	GDP	D80	2	Parallel	Indirect	14-18,25-27
OBG/HflX like superfamily	OBG Family	*B. subtilis* ObgE	1LNZ	2.6	B	G4P	D212	2	Parallel	Indirect	161-165,172-174
		E. coli ObgE	5M04	1.85	A	GDP	D213	2	Parallel	Ambiguous	162-166.A,173-175
TrmE-Era-EngA-YihA-Septin like superfamily		*E. coli* TrmE	2GJ8	1.7	D	GDP + ALFx	D270	2	Parallel	Indirect	219-223,230-232
		B. subtilis EngA	4KYU	1.7	A	GDP	D57	2	Parallel	Indirect	6-10,17-19
	Septins	*S. mansoni* SEPT10	4KVA	2.1	A	GTP	E100	2	Parallel	Direct	44-48, 55-57
		S. mansoni SEPT10	4KV9	1.93	A	GDP	E100	2	Parallel	No Mg^{2+}	44-48, 55-57
Ras-like superfamily	Ras	H-Ras p21	5P21	1.35	A	GNP	D57	2	Parallel	Indirect	6-10,17-19
		H-Ras p21	3KUD	2.15	A	GDP	D57	2	Parallel	Indirect	6-10,17-19
	Rab	Human Rab6A	1YZQ	1.78	A	GNP	D68	2	Parallel	Indirect	16-20,27-29
		Human Rab6A	4DKX	1.9	B	GDP	D68	2	Parallel	Indirect	16-20,27-29
	Rho	Human Cdc42	2QRZ	2.4	B	GCP	D57	2	Parallel	Indirect	6-10,17-19
		Human Cdc42	1A4R	2.5	A	GDP	D57	2	Parallel	Indirect	6-10,17-19
	Ran	Human Ran	1K5D	2.7	J	GNP	D65	2	Parallel	Indirect	13-17,24-26
		Canis lupus Ran	1BYU	2.15	A	GDP	D65	2	Parallel	Indirect	13-17,24-26
	Rag	*S. cerevisae* Gtr2p	3R7W	2.7	B	GNP	E62	2	Parallel	Direct	12-16,23-25
		S. cerevisae Gtr2p	4ARZ	3.1	B	GDP	E62	2	Parallel	No Mg^{2+}	12-16,23-25
	Arf	Human Arf6	2J5X	2.8	A	GSP	D63	2	Parallel	Indirect	16-20,27-29
		Human Arf6	3N5C	1.8	A	GDP	D63	2	Parallel	No Mg^{2+}	16-20,27-29
	Heterotrimeric GTPase G alpha	Human G-alpha-i1	2G83	2.8		GDP + ALFx	D200	2	Parallel	Indirect	36-40,47-49
		Human G-alpha-i1	2OM2	2.2		GDP	D200	2	Parallel	Indirect	36-40,47-49
	MglA	*M. xanthus* MglA	6IZW	2.4	B	GSP	D58	3	Antiparallel	Indirect	15-19,26-28
		M. xanthus MglA	5YMX	1.35	A	GDP	D58	3	Antiparallel	No Mg^{2+}	15-19,26-28
Myosin-kinesin superfamily	Myosin	*D. discoideum* Myosin	1VOM	1.9	A	ADP VO4	D454	2	Parallel	Indirect	174-178,185-187
		D. discoideum	1MMA	2.1	A	ADP	D454	2	Parallel	Indirect	174-178,185-187

(continued on next page)

Table 1 (*continued*)

Superfamily	Family	Protein Name	PDB ID	Res-olution (Å)	Chain ID	Nucleotide	Walker B D/E	Walker B strand number	Walker B strand orientation	Mg^{2+} coordination (Direct/indirect)	P-loop residues (CA superposed)
		Myosin									
Separate Families	Y l q F / Y a w G Family	*T. maritima* YlqF	3CNN	2.3	A	GTP	D30	2	Parallel	No Mg^{2+}	
		T. maritima YlqF	3CNO	2.3	A	GDP	D30	2	Parallel	No Mg^{2+}	
	Dynamin	Human Dynamin	2X2E	2	A	GDP ALF	D136	2	Parallel	Indirect	34-38,45-47
		Human Dynamin	3L43	2.27	B	GDP	D136	2	Parallel	No Mg^{2+}	34-38,45-47
Division: P-loop GTPases; Class: SIMIBI											
MinD/Mrp-ETK superfamily	Mrp/MinD family	*P. furiosus* MinD	1G3R	2.7	A	ACP	D118	2	Parallel	Indirect	6-10,17-19
		P. furiosus MinD	1G3Q	2	A	ADP	D118	2	Parallel	Indirect	4-8,17-19
BioD-FTHFS superfamily	BioD family	*H. pylori* BioD	3QXJ	1.38	A	GTP	E116	2	Parallel	Direct	3-7,14-16
		H. pylori BioD	3QY0	1.6	A	GDP	E116	2	Parallel	Direct	2-6,14-16
Separate families	Signal recognition associated GTPase family	*T. aquaticus* Ffh	2J7P	1.97	B	GNP	D187	2	Parallel	Indirect	101-105,112-114
		T. aquaticus Ffh	1O87	2.1	A	GDP	D187	2	Parallel	Indirect	101-105,112-114
	Clp1 family	*C. elegans* Clp1	4OHV	2.3	A	ANP	N229	2	Parallel	Indirect	116-120,127-129
		C. elegans Clp1	4OI2	2.6	A	ADP	N229	2	Parallel	Indirect	116-120,127-129
	G3E family	*T. kodakarensis* HypA	5AUN	1.8		ACP	D132	2	Parallel	Indirect	23-27,34-36
		T. kodakarensis HypA	5AUO	1.6		ADP	D132	2	Parallel	Indirect	23-27,34-36

Table 2. List of representative structures of P-loop ATPases

Superfamily	Family	Protein name	PDB ID	Res-olution (Å)	Chain ID	Nucleotide	Walker B D/E	Walker B strand number	Walker B strand orientation	Mg2+ coordination (direct/ indirect)	P-loop residues (CA superposed)
Division: P-loop ATPases; class: ASCE											
	PilB	PilB	5TSH	2.3	A	ANP	E357	Other	Other	Indirect	321–325, 332–334
			5TSG	3.4	B	ADP	E357	Other	Other	Indirect	321–325, 332–334
	Terminase	GP2 of bacterial virus Sf6	4IEE	1.9	A	AGS	E118	3	Parallel	Indirect	18–22, 29–31
			4IEI	2.1	A	ADP	E118	3	Parallel	No Mg^{2+}	18–22, 29–31
	F1-ATPase	F1-ATPase	2JDI	1.9	D	ANP	D256	3	Parallel	Indirect	152–156, 163–165
			2V7Q	2.1	D	ADP	D256	3	Parallel	Indirect	152–156, 163–165
	FtsK	FtsK	2IUT	2.3	A	AGS	D595	3	Parallel	Indirect	462–466, 473–475
			2IUU	2.9	A	ADP	D595	3	Parallel	No Mg^{2+}	462–466, 473–475
	ABC transporter	CbiO dimer	5X40	1.5	A	ACP	D165	3	Parallel	Indirect	34–38, 45–47
		MJ1267	1G6H	1.6	A	ADP	D178	3	Parallel	Indirect	36–40, 47–49
	Recombinase	RadA	2FPM	2	A	ANP	D211	3	Parallel	Indirect	101–105, 112–114
		RadA	2FPK	2.1	A	ADP	D211	3	Parallel	Indirect	101–105, 112–114
	SF1	RecD2	3GPL	2.5	A	ANP	D435	3	Parallel	Indirect	356–360, 367–369
		Pif 1 helicase	5FTB	1.38	A	ANP	D105	3	Parallel	Indirect	24–28, 35–37
		Pif 1 helicase	5FTC	2.26	A	ADP	D105	3	Parallel	No Mg^{2+}	24–28, 35–37
	AAA+	ORC2	1W5T	2.4	A	ANP	D145	3	Parallel	Indirect	52–56, 66–68
		ORC2	1W5T	2.4	C	ADP	D145	3	Parallel	Indirect	52–56, 66–68

B residue on strand 3A) is the conformational change that the residue undergoes on GTP binding [19,25,26]. This conformational change facilitates the interaction of MxMglA Asp58 with the water at position 6 (Figure 7(a)). Hence, the conformational change functions as an allosteric regulator of nucleotide binding (Figure 7(a)). This motivated us to search for other examples of P-loop NTPases (representative structures listed in Tables 1 and 2) in which conformational changes in Walker B residue existed between two nucleotide states, GDP/ADP and GTP/ATP bound states, of P-loop NTPases.

In the GTP-bound state of Arf GTPases, the Walker B aspartate forms a water-mediated interaction (through position 6) with the nucleotide-bound Mg^{2+}. In the GDP-bound conformation, the relative movement of strands 2P and 3A results in displacement of the aspartate away from the water molecule at position 6 (Figure 7(b)). SEPTIN10 is an example of septin family of GTPases of the TRAFAC class, where a shift of strand 3A containing the Walker B glutamate was observed on comparison between GDP- and GTP-bound conformations (Figure 7(c)) [35]. A similar translation of Walker B residue between GDP- and GTP-bound subunits was observed in structures of heterodimeric Rag GTPase family members (RagA/RagC and Gtr1p/Gtr2p) (Figure 7(d)). In all these examples, GDP-binding to just the GTPase is not dependent on Mg^{2+} [19,25,26,35,36]. A few of these examples do not possess Mg^{2+} in the GDP-bound structure (e.g., MxMglA), while other structures contain Mg^{2+} in a sub-optimal geometry with the Walker B residues pointing away from the atoms coordinated to Mg^{2+}.

In Arf family, relative movement of strands 2P and 3A with respect to the rest of the central β-sheet between GDP and GTP states results in displacement of an N-terminal amphipathic helix, and acts as an allosteric regulator of binding of Arf GTPases to membranes [36,37]. In the Rag family of GTPases, RagA or RagB exists as a heterodimer with RagC or RagD. In a functional state of the heterodimer, either RagA/RagB or RagC/RagD binds GTP while the other subunit is bound to GDP [38]. Upon GTP binding to one of the subunits, conformational change in α1 helix (Figure 7(d)) of the adjacent protomer is necessary to order to prevent steric hindrance between the two subunits of the dimer [38]. This extension of the α1 helix (Figure 7(d)) leads to shift of strand 2P and displaces the Walker B residue. We propose that the shift in the strands locks the adjacent GTPase subunit in a conformation favorable for GDP binding, but not GTP.

Discussion

Through structural analysis of P-loop NTPases and mutational analysis of MxMglA, we confirmed that Asp58 functions as the Walker B acidic residue

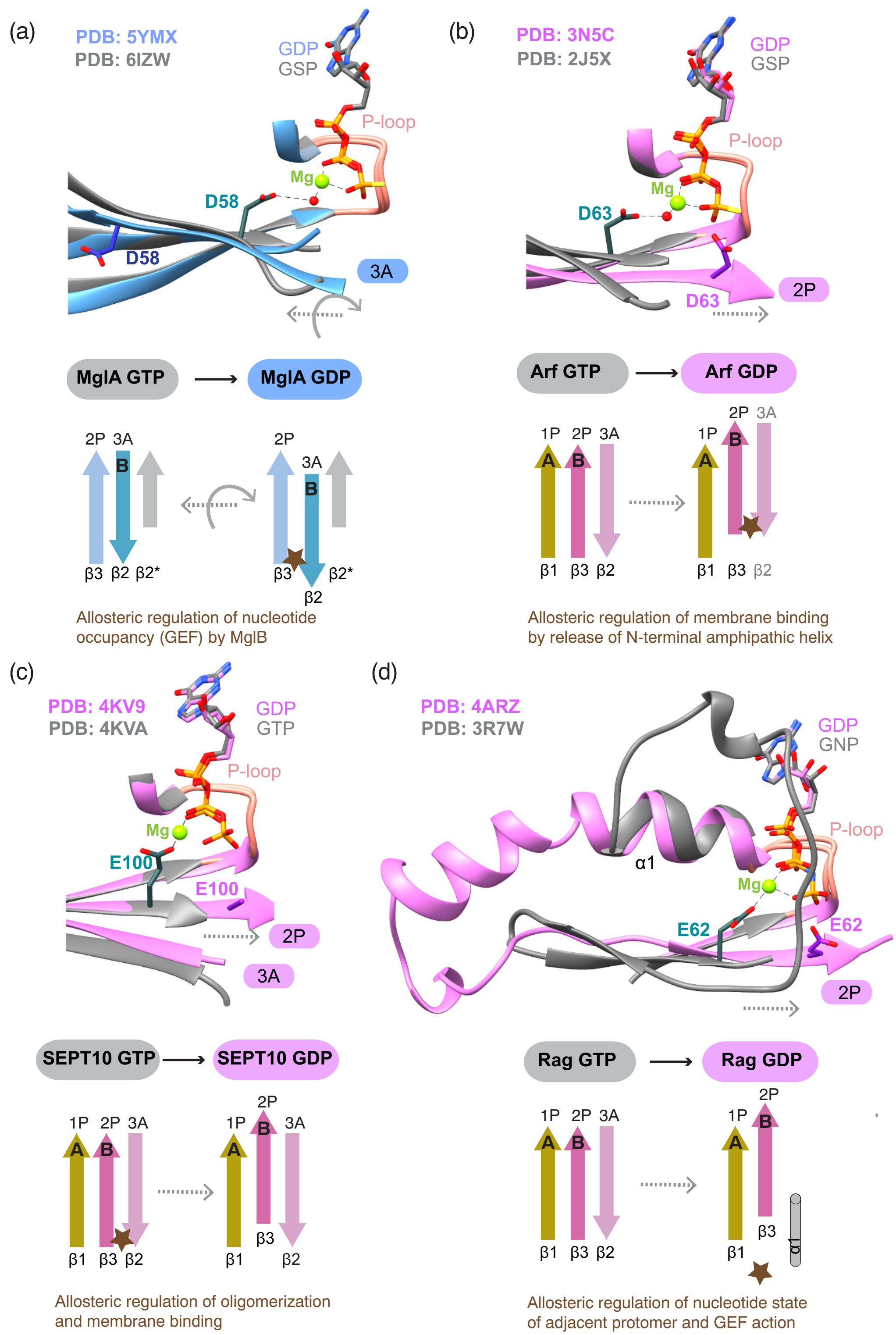


Figure 7. Conformational changes in Walker B upon transition from GTP (active) to GDP (inactive) bound states. Structural superposition of the GTP-equivalent and GDP bound states (a–d), highlighting the movement of Walker B between the two states, and the schematic representation of the movement. (a) MglA-GSP (gray; PDB ID: 6IZW) and MglA-GDP (blue; PDB ID: 5YMX). (b) Arf6-GTP (gray; PDB ID: 2J5X) and Arf6-GDP (pink; PDB ID: 3N5C). (c) SEPT10-GTP (gray; PDB ID: 4KV9) and SEPT10-GDP (pink; PDB ID: 4KVA). E100 labeled in the GDP conformation is truncated till CB atom in the crystal structure. (d) RagC-GTP (gray; PDB ID: 3R7W) and RagC-GDP (pink; PDB ID: 4ARZ).

of MglA family. Our analysis deduced the following features that define a Walker B residue. i) The residue is usually located in strand 2P (parallel in TRAFAC and SIMIBI families) or strand 3 (3P in ASCE and 3A in MglA families) of the core β-sheet. ii) In most P-loop NTPases, the Walker B residue interacts indirectly with the nucleotide-bound Mg^{2+}, mediated by the position 6 water. The spatial orientation of the two major interacting atoms with position 6 water is highly conserved (Figure 4(d)). The two atoms usually belong to the Walker B acidic residue and its adjacent residue, respectively (Figure 5(a)). iii) Hydrogen bond between Walker A threonine/serine and the Walker B acidic residue orients them to coordinate Mg^{2+} (Figure 5(a)). iv) In a few families of ASCE ATPases and in MglA, a threonine/serine present on the adjacent strand contributes to optimal orientation of the Walker B residue and/or position 6 water (Figure 5(b)). v) Based on the structures included in the analysis, the Walker B residue in most P-loop NTPases are either aspartate or glutamate, and very rarely asparagine.

MxMglA Asp58 is part of the $G3^{Alt}$ motif ([L/V][F/A][F/M]**D**) in MglA family of GTPases, a new motif defined in this study. In the structural phylogeny of GTPases, the ancestral prokaryotic families of MglA and Rup have evolved [L/V][F/A][F/M]**D** and **D**xxG motifs, respectively, to facilitate nucleotide binding [20]. It has been proposed that the eukaryotic Arf/Sar/Gα lineage of GTPases arose from MglA family, while the other families of eukaryotic small Ras-like GTPases evolved from Rup [20,21]. Based on our study, we speculate that during evolution of the MglA family into the Arf/Sar lineage, threonine in G3 motif was replaced with aspartate, which gained the function of Mg^{2+} coordination.

The Walker B aspartate in MglA functions as a regulator of nucleotide binding, which is modulated allosterically by MglB. Conformational changes involving Walker B motif regulate catalysis in certain subfamilies of eukaryotic P-loop GTPases, too. Interestingly, many of these subfamilies, such as the Rag family heterodimers [39], and the Arf sub-family [40], are believed to have evolved from the prokaryotic MglA family [22].

Our biochemical study of the MxMglA mutants identifies Thr78 as an important residue for optimal catalytic activity of MxMglA. This observation and the analysis of P-loop NTPase families have led us to identify a probable role played by structurally equivalent threonines in the ASCE family, highlighting a convergent feature of MglA family with that of ASCE family of P-loop ATPases. The occurrence of threonine in certain ASCE family ATPases, which are structurally equivalent to MxMglA Thr78, highlights a convergence in the choice of residues and their spatial conservation for Mg^{2+} coordination through water at position 6. Thus, despite belonging to the TRAFAC class of P-loop GTPases, the MglA family of prokaryotic GTPases possesses two characteristics of the ASCE family ATPases. These include i) presence of Walker B aspartate in strand 3 (strand 3P in ASCE and 3A in MglA) and ii) presence of a conserved threonine in strand 2P.

Based on insights gained from our observations, we conclude that the optimized nucleotide-binding pockets seen in extant members of different P-loop NTPase families have evolved through multiple events of remodeling. The geometry of the interactions, facilitating optimal coordination of Mg^{2+}, has been maintained during the remodeling (Figure 4(d) and 5(a), b), emphasizing its significance in enzyme catalysis. The position of Walker B residue and its interacting motifs across the P-loop NTPase families is an example of plasticity in enzyme families, where spatial equivalence of atoms required for optimal catalysis has been maintained during enzyme evolution [41].

Essentiality of the Walker B aspartate/glutamate has led to conservation of the motif in all families of nucleotide-hydrolyzing enzymes and may be a feature that contributed to the evolution from a nucleotide binding protein to a nucleotide-hydrolyzing enzyme. A recent study of Walker A motifs in proteins led to the design of a minimal nucleotide binding scaffold and a synthetic nucleotide binding protein [42]. The insights obtained on structural and chemical features of Walker B residue and the evolution of P-loop NTPases will extend our ability to design novel proteins that possess a hydrolyzing capability in addition to the ability to merely bind nucleotides.

Methods

Cloning, expression and purification

MxMglA and MxMglB constructs used were as reported [25]. $MxMglA^{D58A}$ and $MxMglA^{T78A}$ mutations were incorporated in MxMglA using restriction free cloning method [43] with the forward primers.

5′-CCGCACGCTCTTCTTCGCCTTCCTGCCGCTGTCGC-3′ for D58A and 5′-CGCTTCCACCTGTACGCGGTGCCCGGTCAGGTC-3′ for T78A mutations. Protein expression and purification protocol for $MxMglA^{D58A}$ and $MxMglA^{T78A}$ were the same as reported for MxMglA [25]. Protocol followed for MxMglB expression and purification was also similar to that reported in Baranwal *et al*. [25].

NADH coupled GTP hydrolysis assay

NADH coupled enzymatic assay, similar to the protocol used in [25], was used to measure GTP hydrolysis activity. A master mix was prepared in A50 buffer (50 mM NaCl, 50 mM Tris (pH 8.0)) containing 600 μM NADH, 1 mM phosphoenol pyruvate, 5 mM $MgCl_2$, 1 mM GTP (all final concentrations) and pyruvate kinase and lactate dehydrogenase enzyme mix (~25 U/ml). All components were mixed in a 200 µl reaction volume at 25 °C, and added to Corning 96 well flat bottom plate. The reactions were initiated by adding MxMglA, MxMglA^{D58A} or MxMglA^{T78A} to a final concentration of 10 µM to the reaction mix, in the presence or absence of equimolar ratio of MxMglB dimer. Decrease in NADH absorbance was measured at 340 nm using a Varioskan Flash multimode plate reader. The absorbance was measured at every 20 s for 7200 s. Initial time point reading and absorbance of buffer components were subtracted from all readings. NADH absorbance was converted to GDP produced using slope obtained from a standard curve containing known concentrations of NADH. GraphPad Prism was used for data analysis and plotting the curves.

Binding assays for MxMglB using fluorescence anisotropy

To estimate binding of MxMglB to nucleotide bound MxMglA mutants, fluorescently-labeled nucleotide *mant*-GDP (m-GDP) or *mant*-GMPPCP (m-GCP) (Jena Bioscience) was used (protocol similar to the one used by Baranwal *et al.* [25]). The change in anisotropy was measured in Fluoromax-4 (Horiba), in a cuvette with 200 µl sample volume at 25 °C, and excitation and emission slit widths of 5-nm bandpass. A mix of 100 nM of *mant*-labeled nucleotide in buffer (50 mM Tris (pH 8.0), 50 mM NaCl and 2 mM $MgCl_2$) with 2 µM of MxMglA (all final concentrations) was titrated with increasing concentrations of MxMglB. Each reading corresponds to 10 averaged single point anisotropy values. The initial value of anisotropy of the mant-labeled nucleotide with MxMglA was subtracted from all the readings. GraphPad Prism was used to plot the values against the concentration of MxMglB dimer (since MxMglA binds to one dimer of MxMglB) and to fit the data to the binding equation for estimation of K_D. The equation used to fit the data points for one site-specific binding is, $y = B_{max} \times x \div (K_D + x)$, where y represents the bound fraction (relative fluorescence anisotropy reading), B_{max} is the anisotropy value at saturation (obtained by fitting the data to the equation), x is the concentration of MxMglB dimer, and K_D is the binding constant. The relative fluorescence anisotropy values were calculated by dividing all the readings by the B_{max} value and subsequently plotted against MxMglB dimer concentration. Each data point is an average of at least three independent measurements, and the standard error is shown in which each data represents three repeats. Error bars represent standard error.

High-pressure liquid chromatography analysis for detecting bound nucleotide

The purified MglA mutants were diluted to 100 µM concentration in buffer A50 and then heat denatured at 75 °C. The heated protein sample was spun at 21,000 ***g*** for 10 min. Then supernatant was filtered with 0.22-µm cellulose acetate filter (Corning), and the sample was loaded on the DNAPac PA200 ion-exchange column (Thermo Fisher). Buffer A (2 mM Tris (pH 8.0)) was used as binding buffer. The runs were performed at a flow rate of 0.4 ml/min, with linear gradients of 0% to 20% Buffer B (2 mM Tris (pH 8.0), 1 M NaCl) for 3 CV (column volumes), 20%–50% Buffer B for 3 CV and 100% B for 2 CV. The sample injection volume was 100 µl. The absorbance at 260 nm, indicative of the presence of GDP, was plotted against conductivity. The GDP standard was run with 100 µl of 500 µM filtered GDP solution.

Nucleotide binding studies using nucleotide exchange kinetics

The kinetic measurements for MxMglA and its mutants with MxMglB were performed on Fluoromax-4 (Horiba) where the intensity of fluorescence emission by *mant*-labeled GDP (Jena Bioscience) at 440 nm was monitored after the excitation at 360 nm (protocol similar to the one used in [25]). The sample volume was 200 µl in a quartz cuvette (10 × 2-mm path length) at 25 °C and excitation and emission slit widths used were 2-nm bandpass. *Mant*-GDP (0.8 µM) was added with Buffer A50 (50 mM Tris, 50 mM NaCl, 5 mM $MgCl_2$ (pH 8.0)). The protein mix, i.e., MxMglA and its mutants (3 µM of MxMglA/mutants and 6 µM MxMglB monomer), was added in the cuvette at 400 s after stabilization of the signal from only *mant*-GDP. Consequently, the fluorescence was recorded for 1400 s where the increase in fluorescence intensity reflected the nucleotide binding kinetics. At 1800 s, the *mant*-labeled nucleotide was competed out with excess of unlabeled GTP (500 µM final concentration), resulting in release of *mant*-labeled nucleotide from the protein which manifested as decrease of fluorescence intensity. For plotting the relative intensities from the measurements, each value was divided by the average of first 200 readings (400 s). These accumulation and decay reactions were fitted to exponential binding equations as given below to estimate the k_{on} and k_{off} values.

$P + N_1 \rightarrow PN_1 + N_2 \rightarrow PN_2 + N_1$, where P represents protein, N_1 is the labeled nucleotide, N_2 is the unlabeled nucleotide, and PN denotes the protein-nucleotide complex.

For estimation of k_{on}, $PN_t = PN_{max}\,(1 - e^{-kt})$.

For estimation of k_{off}, $PN_t = PN_{max} - PN_{min}\,(e^{-kt}) + PN_{min}$.

Here, PN_t represents the amount of the complex at time t, PN_{max} is the maximum amount of the complex upon association, and PN_{min} is the minimum amount of the complex after dissociation.

Size exclusion chromatography

Size exclusion chromatography was used for determining oligomeric status of MxMglA^{D58A} and

MxMglA^{T78A}. Approximately 2–3 mg of protein was injected into a Superdex 75 size exclusion column (GE healthcare). UV absorbance at 280 nm was observed to monitor the elution of the protein in the buffer A50 (50 mM NaCl, 50 mM Tris (pH 8)). The elution volume was compared to that of wild-type MxMglA.

Sequence and structural analysis

Sequences for MglA family were obtained from Wuichet and Søgaard-Andersen [20]. Eukaryotic small Ras-like GTPase sequences were obtained from Wennerberg *et al.* [44]. Sequences were analyzed and visualized in JalView [45]. MUSCLE [46] algorithm was used for alignment for generating Weblogo [47]. List of the P-loop ATPase and GTPase classes was obtained from Refs. [2–4,33] and representative members of families were chosen. The structures for the representative families were chosen based on a search using the protein name in the PDB. Among the list of structures, a pair of structures at the highest resolution with GTP/GDP and ATP/ADP were chosen. The sequences from the PDB were compared with the UniProt entries to ensure that these were not structures of mutants, especially at the catalytic site or Walker B residues. Structural superposition was carried out in Chimera [48] using the "match" command based on Cα atoms of the residues flanking the P-loop, as listed in Tables 1 and 2. 5P21 was used as the reference structure for all superposition steps.

PDB analysis of Mg^{2+} coordinated to nucleotide bound structures

Structures were downloaded from PDB database on 16th December 2019. For this analysis, high-resolution (higher than 3.0 Å) structures, and of file size less than 1 MB were used. Extraction of data from the set of PDB files was done using Python scripts. Structures containing Mg^{2+} were picked and list of Mg^{2+} coordination was generated using the CONECT records in each PDB file. List was generated for Mg^{2+} interacting with any oxygen atom (atom id "O1A," "O2A," "O3A," "O1B," "O2B," "O3B,""O1G," "O2G," "O3G") of the phosphate in nucleotide ligands (Supplementary Table 1), based on the information listed in the "CONECT" records in PDB. Out of these, hexa-coordinated Mg^{2+} was chosen for further analysis. Redundant contacts from multiple chains were excluded. The number of contacts with NTP, amino acid residue and water were estimated from the non-redundant list.

Among the list of structures containing Mg^{2+}, we ensured that two of the contacts were OB and OG atoms, and assigned them as atoms 1 and 2. The rest were temporarily numbered 3 to 6 for calculating 9 angles with Mg^{2+} at the central atom (M) [1-M-2, 1-M-3, 1-M-4, 1-M-5, 1-M-6, 2-M-3, 2-M-4, 2-M-5, 2-M-6]. According to octahedral geometry, two (1-M-3 and 2-M-4) of these should be 180° and the rest 7 should be 90°. Only Mg^{2+} satisfying the criteria where two out of the nine angles were in the range between 150° and 180°, and the rest at 90 (±20)° were chosen for further analysis. Based on the two ~180° angles, positions 3 and 4 were assigned the atoms at 180° from OB and OG, respectively. Positions 5 and 6 were unambiguously assigned based on the sign of the torsion angle between the four atoms 1–2-4-X (where 1, 2 and 4 should lie on one plane and X is either 5 or 6). An atom with positive value of torsion angle (~90°) was assigned position 5, while the other was assigned 6 (~−90°). After assignment of the atom positions, the distances between the coordinating atoms and Mg^{2+} were observed to be within the permissible range of Mg^{2+} coordination features. Structures with sequence identity more than 90%, and with R_{free} higher than 30% were excluded from further analysis using PISCES server [49].

UniProt identifiers were extracted for the final list of structures and submitted to the SUPFAM server [50] in order to distinguish between members belonging to the P-loop family of NTPases and others. ECOD database was used to identify associated F groups [29]. Water-mediated contacts for Mg^{2+} were determined by identifying atoms capable of forming hydrogen bonds within 3.5 Å of the directly coordinating water molecules. These constitute the indirect contacts. The distribution of residues and waters in the direct and indirect contact lists was analyzed and plotted. Angle "θ" and dihedral angle "φ" were defined as shown in Figure 4(a) and calculated using the coordinates of the relevant atoms. "θ" is defined as the angle between Mg^{2+}, the atom at position 6 and the interacting atom of the protein residue in indirect contact (labeled as 6′, Figure 4(a)). "φ" is defined as the dihedral angle between the atom at position 4, Mg^{2+}, water at position 6 and the interacting protein atoms labeled 6′ (Figure 4(a)). The polar plot was obtained using MATLAB.

Figure 3(a) lists the number of structures filtered at each step and Supplementary Tables 2 and 3 list the constituent PDBs of the dataset.

CRediT authorship contribution statement

Manil Kanade: Conceptualization, Methodology, Formal analysis, Investigation, Writing - review & editing, Visualization. **Sukanya Chakraborty:** Methodology, Formal analysis, Investigation, Writing - review & editing, Visualization. **Sanket Satish Shelke:** Methodology, Software, Data curation. **Pananghat Gayathri:** Conceptualization, Methodology, Validation, Formal analysis, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

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Author Contributions

M.K. designed and performed the experiments, sequence and structure analyses and reviewed manuscript writing. S.C. performed structural analysis of GTPases, kinetics and binding assays and reviewed manuscript writing. S.S.S. wrote the script and provided the data for the Mg^{2+} coordination analysis. P.G. conceptualized the study, supervised the experiments and analyses, and wrote the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmb.2020.07.024.

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Abbreviations used:
GAP, GTPase Activating Protein; *mant*, 2′/3′-*O*-(*N*-methyl anthraniloyl; GDP, guanosine di-nucleotide; GNP or GppNHp, guanosine-5′-[(β,γ)-imido]triphosphate; GTP, guanosine 5′-tri-phosphate; GTPγS, guanosine:5′-O-[gamma-thio]triphosphate; PDB, Protein Data Bank; P_i, inorganic phosphate; TtMglA, *T. thermophilus* MglA; TtMglB, *T. thermophilus* MglB; MglA, Mutual gliding motility protein A; MglB, Mutual gliding motility protein B; MxMglA, *M. xanthus* MglA; MxMglB, *M. xanthus* MglB.

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